

Question 1

A Generative AI Engineer must preprocess legal contracts before using them in a retrieval system. What step is most critical for improving retrieval accuracy?

1. Randomly remove redundant sections.

Explanation

Incorrect: Random removal risks excluding critical information.

2. **Chunk the contracts by logical sections, such as clauses or exhibits, and ensure chunks fit within the model's token limits.**

Explanation

Correct: Logical chunking preserves the document's structure and ensures chunks are processable by the model.

3. Retain the entire contract as a single document.

Explanation

Incorrect: Large documents often exceed token limits and hinder efficient processing.

4. Split the contracts into equal-sized chunks.

Explanation

Incorrect: Arbitrary splitting may disrupt meaningful sections and reduce retrieval quality.

Overall explanation

Chunking legal contracts by logical sections ensures the retrieval system maintains context and provides accurate responses.

Question 2

A Generative AI Engineer is creating an LLM-based application where documents for the retriever are chunked to 512 tokens each. The engineer prioritizes cost and latency over quality. Which configuration fulfils their need?

1. **Context length 512; smallest model is 0.13GB and embedding dimension 384.**

Explanation

Correct: This matches the chunk size and uses the smallest model, minimizing both cost and latency.

2. Context length 514; smallest model is 0.44GB and embedding dimension 768.

Explanation

Incorrect: Context length of 514 exceeds the chunk size and adds unnecessary model complexity.

3. Context length 2048; smallest model is 11GB and embedding dimension 2560.

Explanation

Incorrect: Larger context length and embedding size increase latency and cost.

4. Context length 32768; smallest model is 14GB and embedding dimension 4096.

Explanation

Incorrect: This configuration is overly large and does not align with cost or latency priorities.

Overall explanation

A context length of 512 aligns perfectly with the application's chunk size and reduces costs and latency due to the smaller model size.

Question 3

A Generative AI Engineer must deploy a Vector Search index for a knowledge base of scientific papers. The system must support real-time queries with low latency. What combination of tools and resources should they use? **(Choose three)**

1. Lightweight CPU infrastructure to reduce costs.

Explanation

Incorrect: CPUs are insufficient for handling computationally intensive vector search tasks.

2. High-performance GPUs for low-latency computations.

Explanation

Correct: GPUs accelerate vector similarity calculations, enabling fast response times for real-time queries.

3. Distributed databases to store raw documents.

Explanation

Correct: Distributed databases ensure scalable storage for large knowledge bases while separating raw data from indexed vectors.

4. A summarization model trained on scientific datasets.

Explanation

Incorrect: While useful for downstream tasks, a summarizer is not required for the vector index itself.

5. FAISS for creating and querying the vector index.

Explanation

Correct: FAISS provides efficient indexing and querying capabilities, making it ideal for real-time Vector Search.

Overall explanation

Building a Vector Search index requires FAISS for efficient indexing, GPUs for low-latency queries, and distributed databases for scalable document storage.

Question 4

A Generative AI Engineer is designing a conversational AI for customer service. The AI must process queries and recommend actions based on the conversation's context. What should the engineer prioritize in the prompt design?

1. Specify only the tone of the conversation.

Explanation

Incorrect: While tone improves user experience, it does not guide the AI on the required output.

2. Avoid examples to encourage creativity.

Explanation

Incorrect: Examples help the AI understand the task and generate consistent responses.

3. Include clear context and desired output. Example: "User is asking about late deliveries. Recommend next steps to resolve their issue."

Explanation

Correct: Providing context and specifying output ensures relevant and actionable responses based on the conversation.

4. Focus on brevity to generate concise responses.

Explanation

Incorrect: Brevity alone does not ensure relevance or alignment with user needs.

Overall explanation

Providing clear context and expected output guides the model to generate accurate, user-relevant recommendations for customer service interactions.

Question 5

A Generative AI Engineer is tasked with masking sensitive user information in a customer support chatbot. What is the most effective approach to achieve this?

1. Implement regex-based patterns to identify and mask sensitive data like phone numbers and credit card details.

Explanation

Correct: Regex patterns efficiently detect and mask sensitive information, enhancing data privacy.

2. Use a summarization model to exclude sensitive details.

Explanation

Incorrect: Summarization models may not reliably identify all sensitive information.

3. Rely on the LLM's training to avoid generating sensitive outputs.

Explanation

Incorrect: LLMs cannot guarantee the exclusion of sensitive data without explicit preprocessing.

4. Manually filter user inputs before processing.

Explanation

Incorrect: Manual filtering is impractical for large-scale applications.

Overall explanation

Regex-based masking ensures sensitive data is identified and securely masked, maintaining compliance with privacy standards.

Question 6

A Generative AI Engineer is developing a RAG system to answer questions about company financial reports. Users have complained that answers are vague and lack citations. How should the engineer improve the system?

1. Use a summarization model to simplify the financial data.

Explanation

Incorrect: Simplification does not address the issue of vague answers or missing citations.

2. Increase the model temperature to encourage more creative responses.

Explanation

Incorrect: Higher temperature introduces variability but does not improve citation accuracy.

3. Fine-tune the model on financial report datasets.

Explanation

Incorrect: Fine-tuning may improve general relevance but does not inherently solve citation issues.

4. Add a step in the pipeline to include relevant citations with retrieved content.

Explanation

Correct: Including citations ensures responses are transparent and build user trust.

Overall explanation

Adding citations improves transparency and trustworthiness, addressing user concerns about vague answers.

Question 7

A Generative AI Engineer must deploy a Foundation Model API for a conversational agent. The API usage is expected to grow rapidly. How should the engineer optimize for scalability?

1. Use a fixed number of compute resources to handle peak demand.

Explanation

Incorrect: Fixed resources may lead to underutilization or system overload during varying workloads.

2. Implement rate limits without scaling the infrastructure.

Explanation

Incorrect: Rate limits control usage but do not improve scalability.

3. **Implement horizontal scaling with autoscaling policies to handle variable workloads.**

Explanation

Correct: Horizontal scaling with autoscaling policies ensures the system can adapt to growing demand efficiently.

4. Reduce the context length to save computational resources.

Explanation

Incorrect: Reducing context length may degrade response quality without addressing scalability.

Overall explanation

Horizontal scaling ensures the infrastructure adapts to changing workloads while maintaining high performance and reliability.

Question 8

A Generative AI Engineer is processing invoices stored as scanned PDFs for a retrieval-based application. The text includes numerical details like invoice IDs, amounts, and dates. Which Python package is the best fit for extracting this content? **(Choose two)**

1. Tika

Explanation

Tika is a toolkit that provides content extraction capabilities for various file formats, including PDFs. However, Tika does not specialize in OCR-based text extraction from scanned documents like PyTesseract and Textract. Therefore, while Tika can extract content from PDFs, it may not be the most suitable choice for accurately extracting numerical details from scanned PDF invoices.

2. **PyTesseract**

Explanation

PyTesseract is a popular Python package that utilizes the Tesseract OCR engine to extract text from images and scanned documents. It is well-suited for extracting textual content, including numerical details, from scanned PDFs like invoices. Therefore, it is a suitable choice for extracting the content mentioned in the question.

3. **Textract**

Explanation

Textract is a Python package that provides efficient methods for extracting text and structured data from various document formats, including PDFs. It is specifically designed for extracting textual information from scanned documents, making it a suitable choice for processing scanned PDF invoices and extracting numerical details like invoice IDs, amounts, and dates.

4. **PyPDF2**

Explanation

PyPDF2 is a Python library that focuses on reading and manipulating PDF files, but it does not have built-in OCR capabilities like PyTesseract or Textract. While PyPDF2 can be used to extract text from PDFs, it may not be the best fit for extracting textual content from scanned PDFs that require OCR technology for accurate extraction of numerical details.

Overall explanation

PyTesseract and Textract are ideal for extracting numerical and textual details from scanned PDFs, providing accurate OCR-based text retrieval.

Question 9

A Generative AI Engineer needs to evaluate the performance of a RAG system trained on legal documents. What metrics and tools should be used? **(Choose two)**

1. BLEU scores for evaluating output text similarity.

Explanation

Incorrect: BLEU is used for text generation tasks, not document retrieval.

2. Token usage to track computational costs.

Explanation

Incorrect: Token usage does not directly measure retrieval performance.

3. **Recall to measure how many relevant documents are retrieved.**

Explanation

Correct: Recall is critical for evaluating the system's ability to retrieve all relevant documents.

4. **NDCG (Normalized Discounted Cumulative Gain) to measure ranking quality.**

Explanation

Correct: NDCG evaluates the relevance and ranking order of retrieved documents.

5. Latency to measure query processing speed.

Explanation

Incorrect: Latency measures response time but not retrieval relevance.

Overall explanation

Combining recall and NDCG ensures the system retrieves relevant documents efficiently and ranks them accurately.

Question 10

A Generative AI Engineer is working on a chatbot that summarizes customer queries for a support team. The chatbot occasionally leaks sensitive data in summaries. What guardrail technique is most effective?

1. Use postprocessing filters to detect and redact sensitive information from generated outputs.

Explanation

Correct: Postprocessing ensures sensitive data is removed from summaries before they are shared.

2. Manually review all summaries for sensitive data before sharing.

Explanation

Incorrect: Manual review is impractical for large-scale applications.

3. Reduce the model's temperature to limit variability in summaries.

Explanation

Incorrect: Temperature adjustments do not address the risk of sensitive data leakage.

4. Rely on the chatbot's training to avoid sensitive data leakage.

Explanation

Incorrect: Training alone cannot guarantee the exclusion of sensitive information.

Overall explanation

Postprocessing filters ensure sensitive data is identified and redacted from summaries, maintaining user trust and compliance.

Question 11

A Generative AI Engineer is tasked with deploying an LLM application that uses a vector store for document retrieval. The application must ensure low latency and scalability. What infrastructure should they prioritize?

1. Skip using a vector store and rely on direct LLM query processing.

Explanation

Incorrect: Without a vector store, retrieval quality and latency will degrade.

2. Deploy the application on a single-node server to minimize complexity.

Explanation

Incorrect: Single-node servers lack scalability for high-demand applications.

3. Deploy the application on a scalable cloud infrastructure with a high-performance database for the vector store.

Explanation

Correct: Cloud scalability ensures the application can handle growing workloads, while high-performance databases reduce latency for vector queries.

4. Use local storage for vector data to minimize costs.

Explanation

Incorrect: Local storage may reduce costs but compromises scalability and latency for production systems.

Overall explanation

Scalable cloud infrastructure with a high-performance vector store ensures low latency and efficient scaling for production-grade systems.

Question 12

A Generative AI Engineer is deploying a model that requires both embeddings and an LLM for document summarization. The model must handle high throughput for real-time queries. What infrastructure should they prioritize?

1. Deploy on a distributed cloud infrastructure with GPU acceleration for model inference and storage optimization for embeddings.

Explanation

Correct: Distributed cloud infrastructure ensures scalability, while GPU acceleration improves inference speed and storage optimization enhances retrieval performance.

2. Use a local server with CPU-only resources to save costs.

Explanation

Incorrect: CPU-only resources cannot handle high-throughput requirements effectively.

3. Deploy the model on a single cloud instance with minimal scaling capabilities.

Explanation

Incorrect: Minimal scaling is insufficient for real-time, high-throughput demands.

4. Skip embedding storage and rely only on the LLM for summarization.

Explanation

Incorrect: Embedding storage is crucial for semantic understanding and efficient retrieval.

Overall explanation

Distributed cloud infrastructure with GPUs and optimized storage ensures scalability and performance for real-time summarization tasks.

Question 13

A Generative AI Engineer is deploying a **pyfunc** model that predicts loan approvals. The model requires pre- and post-processing. What steps should the engineer take to ensure usability?

1. Pass raw inputs directly to the model for faster predictions.

Explanation

Incorrect: Preprocessing ensures data quality and improves prediction reliability.

2. Preprocess inputs (e.g., clean and standardize data), run predictions through the pyfunc model, and format outputs (e.g., "Approved" or "Denied") for user clarity.

Explanation

Correct: Pre- and post-processing ensure the model processes inputs effectively and presents outputs in a user-friendly format.

3. Use only preprocessing without any post-processing.

Explanation

Incorrect: Post-processing is essential for making model outputs interpretable.

4. Skip post-processing and return raw model scores.

Explanation

Incorrect: Raw scores may confuse users and lack actionable insights.

Overall explanation

A workflow combining preprocessing, pyfunc predictions, and post-processing ensures accurate and user-friendly loan approval results.

Question 14

How can metaprompts help in reducing private data exposure in an LLM application?

1. Metaprompts can instruct the model to avoid including specific types of sensitive information in its outputs.

Explanation

Correct: Metaprompts guide the model to recognize and avoid generating outputs containing private data.

2. Metaprompts adjust the temperature for less risky responses.

Explanation

Incorrect: Temperature adjustments do not directly address private data exposure.

3. Metaprompts block all numerical outputs.

Explanation

Incorrect: Blocking numerical outputs is overly restrictive and not related to privacy protection.

4. Metaprompts reduce response length, minimizing information exposure.

Explanation

Incorrect: Reducing response length does not prevent sensitive information leakage.

Overall explanation

Metaprompts help guide LLMs to avoid exposing private data by explicitly defining rules or safeguards in the instructions.

Question 15

A Generative AI Engineer must select an embedding model for FAQ retrieval. The system must support fast lookups and high accuracy. What strategy ensures optimal performance? **(Choose two)**

1. Use a small embedding model to save costs.

Explanation

Incorrect: Small models may compromise accuracy, reducing overall performance.

2. Prioritize token length over speed.

Explanation

Incorrect: Token length is irrelevant for short FAQs.

3. Choose an embedding model with high accuracy on semantic similarity benchmarks.

Explanation

Correct: High similarity accuracy ensures that FAQs and queries are aligned effectively.

4. Use a generative model for response generation instead of embeddings.

Explanation

Incorrect: Generative models do not directly optimize retrieval accuracy.

5. Use approximate nearest neighbor (ANN) search for faster lookups.

Explanation

Correct: ANN search balances speed and accuracy, making it ideal for large datasets.

Overall explanation

High semantic similarity accuracy combined with ANN search ensures fast and accurate retrieval in FAQ systems.

Question 16

A Generative AI Engineer is tasked with selecting an embedding model for a recommendation system. The system must process product descriptions and customer reviews. What factors are most critical for selecting the embedding model?

1. Context length and semantic understanding of product descriptions and reviews.

Explanation

Correct: Embedding models with longer context lengths and strong semantic understanding perform better for complex inputs.

2. Compatibility with pre-trained LLMs.

Explanation

Incorrect: Compatibility is not as critical as the embedding model's ability to process relevant inputs.

3. Availability of public pre-trained embeddings.

Explanation

Incorrect: Pre-trained embeddings must also align with domain-specific requirements.

4. Tokenization efficiency to reduce computational costs.

Explanation

Incorrect: While important, tokenization does not directly affect semantic understanding.

Overall explanation

Embedding models with long context lengths and robust semantic understanding improve the system's ability to align reviews with product descriptions.

Question 17

A Generative AI Engineer is responsible for developing a chatbot to enable their company's internal HelpDesk Call Center team to more quickly find related tickets and provide resolution. While creating the GenAI application work breakdown tasks for this project, they realize they need to start planning which data sources (either Unity Catalog volume or Delta table) they could choose for this application. They have collected several candidate data sources for consideration: call_rep_history: a Delta table with primary keys representative_id, call_id. This table is maintained to calculate representatives' call resolution from fields call_duration and call_start_time. transcript Volume: a Unity Catalog Volume of all recordings as *.wav files, but also a text

transcript as *.txt files. call_cust_history: a Delta table with primary keys customer_id, call_id. This table is maintained to calculate how much internal customers use the HelpDesk to make sure that the charge-back model is consistent with actual service use. call_detail: a Delta table that includes a snapshot of all call details updated hourly. It includes root_cause and resolution fields, but those fields may be empty for calls that are still active. maintenance_schedule: a Delta table that includes a listing of both HelpDesk application outages as well as planned upcoming maintenance downtimes. They need sources that could add context to best identify ticket root cause and resolution. Which TWO sources do that? **(Choose two)**

1. call_cust_history

Explanation

Incorrect: Customer usage data focuses on tracking service usage but does not aid in identifying ticket root causes or resolutions.

2. maintenance_schedule

Explanation

Incorrect: While it tracks outages and planned downtimes, this information does not directly help identify specific ticket root causes or resolutions.

3. transcript Volume

Explanation

Correct: Text transcripts provide detailed and contextual insights into calls, helping to identify root causes and resolutions.

4. call_detail

Explanation

Correct: This table contains root_cause and resolution fields, making it a primary source for identifying ticket resolutions.

5. call_rep_history

Explanation

Incorrect: This table tracks representative performance metrics and does not provide resolution context.

Overall explanation

Using call_detail for its structured fields (root_cause and resolution) and transcript Volume for detailed call context ensures the chatbot has comprehensive data to resolve tickets efficiently.

Question 18

A Generative AI Engineer is selecting source documents for training a RAG system that answers healthcare questions. What criteria should guide document selection?

1. Include all publicly available medical texts regardless of relevance.

Explanation

Incorrect: Irrelevant texts dilute the system's accuracy.

2. **Select documents with high-quality medical information relevant to the target domain.**

Explanation

Correct: High-quality, relevant documents ensure reliable and accurate system responses.

3. Prioritize lengthy documents to maximize training data size.

Explanation

Incorrect: Length alone does not ensure relevance or quality.

4. Select documents with diverse but non-specialized medical content.

Explanation

Incorrect: General content may lack the specificity required for healthcare applications.

Overall explanation

Choosing high-quality, relevant medical documents ensures the RAG system retrieves accurate answers aligned with healthcare queries.

Question 19

A Generative AI Engineer is tasked with choosing an embedding model for search engine handling long research papers. What factor is most critical? **(Choose two)**

1. Training dataset size.

Explanation

Incorrect: Dataset size is secondary to task-specific evaluation metrics.

2. Token diversity.

Explanation

Incorrect: Token diversity is not a metric relevant to embeddings.

3. **Semantic similarity accuracy.**

Explanation

Correct: High accuracy ensures that search queries retrieve the most relevant results.

4. Cost of inference.

Explanation

Incorrect: While important, cost does not directly affect model performance for long documents.

5. **Context length supported by the embedding model.**

Explanation

Correct: Long context lengths allow better representation of lengthy research papers.

Overall explanation

Long context support and high semantic similarity accuracy ensure effective embeddings for handling long research papers.

Question 20

A Generative AI Engineer must deploy an LLM application that uses Foundation Model APIs. The application needs to handle document summarization and integrate retrieval capabilities. What sequence should they follow to deploy the application effectively?

1. Deploy the endpoint first and configure the embedding model later.

Explanation

Incorrect: Embeddings must be configured before deployment for seamless functionality.

2. Skip retrieval capabilities and rely solely on the LLM for summarization.

Explanation

Incorrect: Retrieval capabilities enhance the accuracy and relevance of summaries.

3. Use pre-trained summarization models without retrieval support.

Explanation

Incorrect: Pre-trained summarization models lack dynamic retrieval capabilities required for the application.

4. Skip embedding creation and rely on document metadata.

Explanation

Incorrect: Embedding creation is critical for vector search and semantic retrieval.

5. **Configure Foundation Model API access, create embeddings for documents, build a vector store, and deploy the endpoint.**

Explanation

Correct: This sequence ensures that document retrieval and summarization are integrated effectively with the LLM.

Overall explanation

Configuring APIs, embeddings, and vector stores ensures the LLM application handles summarization and retrieval seamlessly.

Question 21

A Generative AI Engineer must deploy an LLM application using Foundation Model APIs for real-time customer support. What considerations are necessary for optimal serving?

1. Use Foundation Model APIs without monitoring latency or compute resources.

Explanation

Incorrect: Neglecting resource management leads to potential performance bottlenecks.

2. Skip API rate limits to allow unrestricted queries.

Explanation

Incorrect: Unrestricted queries can overwhelm the system, reducing reliability.

3. Reduce compute resources to minimize costs.

Explanation

Incorrect: Reducing resources may degrade performance and response times.

4. Ensure sufficient compute resources, manage API rate limits, and monitor latency.

Explanation

Correct: These considerations ensure efficient, reliable, and responsive application performance.

Overall explanation

Managing compute resources, API rate limits, and latency ensures optimal performance and user satisfaction in real-time applications.

Question 22

A Generative AI Engineer is tasked with monitoring a production LLM application for a multilingual customer support system. What key metrics should they track? **(Choose two)**

1. BLEU scores for multilingual responses.

Explanation

Incorrect: BLEU evaluates language generation quality but is not critical for overall monitoring.

2. Fine-tuning accuracy for each language.

Explanation

Incorrect: Fine-tuning metrics are relevant for training, not deployment monitoring.

3. Token usage and cost per query.

Explanation

Correct: Tracking token usage and costs helps optimize the system's efficiency in production.

4. Latency, response accuracy, and language coverage.

Explanation

Correct: These metrics ensure responsiveness, correctness, and the system's ability to handle multiple languages effectively.

5. Input-output length ratio.

Explanation

Incorrect: Input-output length ratio does not provide direct insights into application performance or language coverage.

Overall explanation

Tracking latency, accuracy, language coverage, and costs ensures comprehensive monitoring of a multilingual customer support LLM.

Question 23

A Generative AI Engineer must evaluate the success of a summarization pipeline for legal documents. The generated summaries must be concise and include all critical legal points. Which metrics should the engineer prioritize?

1. BLEU and response length to measure accuracy and conciseness.

Explanation

Incorrect: BLEU measures similarity, but response length alone does not ensure content relevance.

2. Latency and perplexity to track speed and fluency.

Explanation

Incorrect: Latency and perplexity do not directly evaluate content relevance or coverage.

3. ROUGE and recall to measure content relevance and coverage.

Explanation

Correct: ROUGE assesses overlap with reference summaries, while recall ensures critical points are included.

4. Precision and retrieval accuracy to optimize inputs.

Explanation

Incorrect: Precision and retrieval metrics do not evaluate summary quality.

Overall explanation

ROUGE and recall ensure the summaries are concise, relevant, and comprehensive, meeting the requirements for legal document analysis.

Question 24

A Generative AI Engineer is asked to deploy a pipeline that retrieves relevant documents and answers user queries. Which component is critical for improving retrieval accuracy?

1. Shorter document chunk lengths.

Explanation

Incorrect: While helpful, chunk length is secondary to embedding quality.

2. A high-quality embedding model trained on domain-specific data.

Explanation

Correct: Domain-specific embedding models improve the accuracy of document retrieval by aligning embeddings with user queries.

3. Pre-trained response generation templates.

Explanation

Incorrect: Response generation templates do not directly impact retrieval accuracy.

4. A fine-tuned classification model.

Explanation

Incorrect: Classification models do not enhance retrieval accuracy in this context.

Overall explanation

Using a domain-specific embedding model ensures that embeddings align with both documents and queries, improving retrieval accuracy.

Question 25

A Generative AI Engineer must select a model from a marketplace for generating personalized financial advice. The system must prioritize accuracy and compliance. What attributes of the model should guide the selection? **(Choose two)**

1. Select a model with low perplexity to ensure fluency.

Explanation

Incorrect: Fluency is secondary to domain accuracy and compliance.

2. Focus on models with larger parameter counts for better performance.

Explanation

Incorrect: Parameter count alone does not guarantee domain relevance or accuracy.

3. Choose a model with low latency for real-time responses.

Explanation

Incorrect: Latency is less critical for compliance and accuracy.

4. Evaluate the model's metadata for training domain and fine-tuning details.

Explanation

Correct: Metadata provides insights into the model's suitability for financial tasks.

5. Prioritize models with high accuracy on financial datasets.

Explanation

Correct: High accuracy ensures reliability and alignment with user needs.

Overall explanation

Reviewing metadata and prioritizing domain-specific accuracy ensures the selected model meets the requirements for personalized financial advice.

Question 26

A Generative AI Engineer must preprocess multilingual documents for a RAG system that serves English-only queries. What preprocessing step ensures consistency?

1. Translate all non-English content into English before further processing.

Explanation

Incorrect: Translation introduces additional overhead and potential inaccuracies.

2. Process all documents without identifying their language.

Explanation

Incorrect: Ignoring language consistency risks irrelevant or incorrect outputs.

3. Retain all documents to provide potential multilingual functionality.

Explanation

Incorrect: Non-English documents may interfere with the quality of English-only responses.

4. Detect and filter non-English documents using a language detection tool.

Explanation

Correct: Filtering ensures only English documents remain, aligning the dataset with the system's requirements.

Overall explanation

Filtering non-English documents ensures the dataset is aligned with the system's language requirements, avoiding irrelevant responses.

Question 27

A Generative AI Engineer needs to design a prompt for a document summarization system that highlights key points and outputs summaries in bullet points. What should the prompt include?

1. Create an abstract of the document for a general overview.

Explanation

Incorrect: Abstracts are not specific to key points or bullet formatting.

2. Summarize the document in bullet points highlighting the key points.

Explanation

Correct: This prompt explicitly defines the output format and key elements to include, ensuring precise results.

3. Generate a summary of the document.

Explanation

Incorrect: This instruction lacks specificity regarding the bullet point format.

4. Highlight the main ideas of the document in a concise paragraph.

Explanation

Incorrect: This instruction may lead to paragraph-style summaries rather than bullet points.

Overall explanation

Explicit prompts defining both the format (bullet points) and content requirements ensure that the summarization system outputs aligned results.

Question 28

A Generative AI Engineer is tasked with designing a system that converts long academic papers into concise summaries for students. The system should highlight key arguments and conclusions. What LLM configuration is most effective?

1. Reduce the model's response length to ensure conciseness.

Explanation

Incorrect: Reducing response length does not guarantee meaningful summaries.

2. Use a generic LLM without fine-tuning.

Explanation

Incorrect: Generic models may generate summaries that lack domain-specific insights.

3. Increase the model's temperature to produce more diverse summaries.

Explanation

Incorrect: Higher temperature introduces randomness, which can reduce summary accuracy.

4. Fine-tune a summarization LLM on academic texts to generate concise summaries.

Explanation

Correct: Fine-tuning on domain-specific data ensures accurate and relevant summaries tailored to academic content.

Overall explanation

Fine-tuning a summarization LLM ensures that the generated summaries capture key arguments and conclusions accurately.

Question 29

A Generative AI Engineer is tasked with controlling LLM costs for a RAG application deployed on Databricks. What strategies should they implement? (Choose two)

1. Optimize token usage to minimize input-output lengths for each query.

Explanation

Correct: Reducing token usage directly lowers inference costs.

2. Increase context length for more comprehensive outputs.

Explanation

Incorrect: Longer context increases token usage, raising costs.

3. Rely on fine-tuning the largest model for all queries.

Explanation

Incorrect: Fine-tuning large models for all tasks is cost-prohibitive and inefficient.

4. Limit API calls by caching frequently retrieved outputs.

Explanation

Incorrect: Caching outputs reduces latency but does not significantly impact LLM inference costs.

5. Use smaller LLMs for non-complex queries while reserving larger models for critical tasks.

Explanation

Correct: Selecting appropriate model sizes based on query complexity optimizes cost-efficiency.

Overall explanation

Optimizing token usage and selecting models based on task complexity are effective strategies for managing LLM costs in Databricks.

Question 30

What is the primary purpose of guardrails in Generative AI applications?

1. To limit model outputs to a fixed length.

Explanation

Incorrect: Guardrails are not primarily designed for length control.

2. To reduce latency in model inference.

Explanation

Incorrect: Guardrails do not directly address performance optimization like latency.

3. To improve response diversity for creative tasks.

Explanation

Incorrect: Guardrails focus on safety and accuracy, not response diversity.

4. To ensure outputs are safe, accurate, and aligned with application requirements.

Explanation

Correct: Guardrails help mitigate risks such as hallucinations, toxic responses, and privacy breaches.

Overall explanation

Guardrails ensure that Generative AI applications produce outputs that are safe, ethical, and meet specific quality requirements.

Question 31

A Generative AI Engineer is tasked with training a chatbot to summarize long technical manuals. Many manuals contain redundant examples that inflate the dataset size. What should the engineer prioritize during preprocessing?

1. Remove redundant examples while preserving unique, meaningful content.

Explanation

Correct: Removing redundancy reduces dataset size and ensures the model focuses on unique content, improving summary quality.

2. Exclude entire manuals containing redundancy.

Explanation

Incorrect: Excluding entire manuals risks removing valuable content alongside redundancy.

3. Retain all examples for completeness.

Explanation

Incorrect: Including redundant examples unnecessarily inflates the dataset without adding value.

4. Replace redundant examples with placeholders.

Explanation

Incorrect: Replacing content with placeholders does not add clarity and may confuse the model.

Overall explanation

Eliminating redundant examples ensures concise, high-quality input, which improves summarization model training.

Question 32

A Generative AI Engineer must augment prompts dynamically based on user input. The user queries a customer support chatbot about "late delivery." What is the best way to handle this?

1. Use a generic prompt like "Explain the issue."

Explanation

Incorrect: A generic prompt lacks context, reducing response accuracy.

2. Exclude all contextual information to keep the prompt concise.

Explanation

Incorrect: Excluding context risks generating irrelevant or unhelpful responses.

3. Include relevant context such as the order date, ID, and expected delivery time in the prompt.

Explanation

Correct: Adding key details helps the LLM generate a precise and relevant response.

4. Rely only on pre-written responses.

Explanation

Incorrect: Pre-written responses cannot dynamically adapt to specific user inputs.

Overall explanation

Augmenting prompts with relevant context ensures the LLM generates precise and user-specific responses.

Question 33

A Generative AI Engineer must select a chunking strategy for summarizing government reports with 10,000+ words. What is the best approach?

1. Process the entire document as a single chunk.

Explanation

Incorrect: Processing a long document as one chunk exceeds most model token limits.

2. Overlap all chunks by 200 tokens.

Explanation

Incorrect: Overlapping improves context but does not align with logical document sections.

3. Chunk based on sections like "Introduction" and "Findings," ensuring each chunk stays within token limits.

Explanation

Correct: Section-based chunking maintains coherence and relevance, aligning with the document structure.

4. Split the document into equal-sized chunks of 1,000 tokens.

Explanation

Incorrect: Arbitrary splitting risks breaking logical sections, reducing summarization quality.

Overall explanation

Chunking by sections preserves the logical flow and ensures the model processes content effectively within token constraints.

Question 34

A Generative AI Engineer must select an LLM for an FAQ retrieval system. The system must balance accuracy, cost efficiency, and scalability. Which criteria should guide the selection process?

(Choose three)

1. Response perplexity to ensure fluent and natural outputs.

Explanation

Correct: Low perplexity indicates the model generates coherent and readable responses, enhancing user satisfaction.

2. Latency metrics to ensure real-time responsiveness for user queries.

Explanation

Correct: Low latency ensures the system is responsive and user-friendly.

3. Model size and architecture to control computational costs.

Explanation

Correct: Smaller, optimized models can reduce inference costs while maintaining scalability.

4. Training dataset diversity for better generalization.

Explanation

Incorrect: Training data affects model generalization but does not directly influence deployment costs or scalability.

5. Token usage per query for cost tracking.

Explanation

Incorrect: Token usage impacts cost but does not directly reflect model accuracy or scalability.

Overall explanation

Balancing latency, model size, and perplexity ensures the FAQ system is accurate, cost-effective, and scalable.

Question 35

A Generative AI Engineer must deploy a basic RAG application for a customer support system. The application requires embedding models for vector search, retrievers for document lookup, and an LLM for response generation. What are the steps to deploy this application?

1. Only configure the retriever and directly connect it to the LLM for responses.

Explanation

Incorrect: Retrievers alone cannot handle vector search or embeddings, limiting retrieval quality.

2. Install dependencies, prepare embedding models, configure retrievers, and deploy an endpoint using model serving.

Explanation

Correct: This sequence ensures that all components, from embedding to deployment, are correctly configured and integrated.

3. Deploy the LLM first, then configure retrievers and embeddings post-deployment.

Explanation

Incorrect: LLM deployment must follow proper integration of embeddings and retrievers for effective responses.

4. Use an embedding model for storage and retrieval, skipping retrievers entirely.

Explanation

Incorrect: Retrievers are essential for precise document lookup in RAG systems.

Overall explanation

A proper sequence, including embedding, retrievers, and endpoint deployment, ensures a functional and optimized RAG application.

Question 36

A Generative AI Engineer is tasked with designing a pipeline to summarize legal documents accurately. How should they evaluate the effectiveness of the summarization model?

1. Measure response latency only.

Explanation

Incorrect: Latency does not directly indicate summarization quality.

2. Use token counts to measure the length of the summaries.

Explanation

Incorrect: Token counts do not evaluate the accuracy or relevance of generated summaries.

3. Focus solely on user feedback.

Explanation

Incorrect: User feedback is helpful but subjective and not sufficient for model evaluation.

4. Use metrics such as ROUGE and BLEU scores to measure summarization quality.

Explanation

Correct: ROUGE and BLEU evaluate the accuracy and relevance of summaries compared to reference texts.

Overall explanation

ROUGE and BLEU are widely used metrics for evaluating the accuracy and quality of summarization models, ensuring reliable performance.

Question 37

A Generative AI Engineer must handle biased data in a RAG application that delivers hiring recommendations. What is the best approach to address bias in the dataset?

1. Ignore the bias, as the model generates outputs based on user queries.

Explanation

Incorrect: Ignoring bias risks perpetuating unfairness in the system.

2. Analyze the dataset for potential biases and remove or reweight biased entries to ensure fairness.

Explanation

Correct: Identifying and addressing biases during preprocessing ensures the model provides equitable and unbiased outputs.

3. Use only pre-trained models and avoid modifying the dataset.

Explanation

Incorrect: Pre-trained models may still reflect the biases present in the original dataset.

4. Reduce the dataset size to exclude potentially biased information.

Explanation

Incorrect: Simply reducing dataset size does not guarantee the removal of biases.

Overall explanation

Analyzing and addressing biases in the dataset ensures fairness in the application and reduces the risk of perpetuating discrimination.

Question 38

A Generative AI Engineer is tasked with summarizing user reviews for a product. The summary must highlight common themes and sentiment. What is the most suitable model task?

1. Named entity recognition

Explanation

Incorrect: NER identifies entities like names or locations but does not summarize reviews.

2. Sentiment analysis

Explanation

Incorrect: Sentiment analysis identifies emotional tone but does not summarize common themes.

3. **Text summarization**

Explanation

Correct: Summarization condenses user reviews into a concise overview, capturing common themes and sentiment.

4. Text classification

Explanation

Incorrect: Classification assigns reviews to categories but does not summarize their content.

Overall explanation

Text summarization effectively captures the essence of user reviews, providing concise insights into themes and sentiment.

Question 39

A Generative AI Engineer is tasked with developing a multi-stage reasoning pipeline for a customer support chatbot. The chatbot must identify intent, retrieve relevant documents, and generate detailed responses. What is a critical design principle for the pipeline?

1. Focus on generating responses without retrieval.

Explanation

Incorrect: Retrieval is essential for grounding responses in relevant documents.

2. Use rule-based systems for document retrieval.

Explanation

Incorrect: Rule-based systems are rigid and may not adapt to diverse queries effectively.

3. Ensure tools are interoperable and can share data seamlessly between stages.

Explanation

Correct: Interoperable tools enable smooth transitions between intent identification, retrieval, and response generation stages, avoiding bottlenecks.

4. Minimize the number of tools to reduce complexity.

Explanation

Incorrect: Reducing tools can limit the chatbot's ability to handle complex tasks requiring multi-stage reasoning.

Overall explanation

Multi-stage reasoning systems rely on interoperable tools to enable efficient workflows, ensuring smooth transitions between tasks like intent identification, retrieval, and response generation.

Question 40

A Generative AI Engineer is creating a pre- and post-processing pipeline for a PyFunc model. Why is this approach useful?

1. It reduces the need for retraining the model.

Explanation

Incorrect: Pre- and post-processing pipelines do not replace retraining for model improvements.

2. It minimizes API costs by batching inputs.

Explanation

Incorrect: API cost optimization is unrelated to pre- and post-processing logic.

3. It automates hyperparameter tuning for the model.

Explanation

Incorrect: Hyperparameter tuning requires separate optimization processes.

4. It allows data to be transformed before and after model inference, ensuring compatibility and optimized outputs.

Explanation

Correct: Pre- and post-processing pipelines handle input normalization and output formatting, enhancing system performance.

Overall explanation

Pre- and post-processing pipelines streamline workflows by transforming inputs and outputs, improving compatibility and system efficiency.

Question 41

A Generative AI Engineer must evaluate a new LLM for summarizing research papers. The evaluation should prioritize accuracy and informativeness. What metrics and tools are most appropriate?

(Choose two)

1. Perplexity for evaluating the fluency of summaries.

Explanation

Incorrect: Perplexity assesses fluency but does not measure informativeness.

2. Token usage to evaluate computational cost.

Explanation

Incorrect: Token usage is unrelated to accuracy or informativeness.

3. ROUGE for measuring content overlap between generated summaries and reference summaries.

Explanation

Correct: ROUGE evaluates the quality of content overlap effectively.

4. Latency to measure the model's response speed.

Explanation

Incorrect: Latency measures performance but not summary quality.

5. Human qualitative assessment for relevance and informativeness.

Explanation

Correct: Human evaluation provides nuanced insights into summary quality.

Overall explanation

Combining ROUGE with human qualitative assessments ensures the evaluation captures both quantitative and qualitative aspects of summary quality.

Question 42

A Generative AI Engineer must select chain components for a system that answers user queries about company policies and cites specific policy sections. What components are essential?

1. A decision tree for guiding users to policy answers.

Explanation

Incorrect: Decision trees are rigid and do not support dynamic queries or citations.

2. A retriever to fetch relevant policy sections and an LLM to generate responses with citations.

Explanation

Correct: Retrievers ensure relevant content is retrieved, and LLMs dynamically generate accurate answers with citations.

3. A classification model to categorize user queries by policy topic.

Explanation

Incorrect: Classification does not address the need for citations or detailed answers.

4. A summarization model to condense policy documents.

Explanation

Incorrect: Summarization lacks the specificity required for citations or query handling.

Overall explanation

Combining retrievers and LLMs ensures accurate query responses with properly cited policy references.

Question 43

A Generative AI Engineer is tasked with selecting a chunking strategy for legal documents to optimize retrieval in a RAG system. The documents include lengthy paragraphs and footnotes. What strategy should they adopt?

1. Split the document into equal-sized chunks without considering content structure.

Explanation

Incorrect: Equal-sized chunks risk breaking meaningful sections, reducing relevance.

2. Overlap all chunks by 200 tokens.

Explanation

Incorrect: Overlapping creates redundancy without necessarily improving retrieval performance.

3. Divide the document into logical sections based on paragraphs, ensuring chunks remain within token limits.

Explanation

Correct: Logical chunking maintains coherence while staying within token constraints, improving retrieval quality.

4. Process the entire document as a single chunk.

Explanation

Incorrect: Single chunks often exceed token limits, making them infeasible for processing.

Overall explanation

Logical chunking ensures that sections remain meaningful and fit within token limits, optimizing retrieval results.

Question 44

A Generative AI Engineer is building a QA system to assist medical professionals. The system must retrieve answers from medical guidelines and provide explanations. What chain components are required?

1. A classification model to categorize queries by topic.

Explanation

Incorrect: Classification does not provide detailed responses or explanations.

2. A summarization model to condense guideline content.

Explanation

Incorrect: Summarization lacks the ability to answer specific questions.

3. A retriever to fetch guideline sections and an LLM to generate responses with explanations.

Explanation

Correct: Retrievers ensure relevant content retrieval, and LLMs create contextually accurate explanations.

4. A rule-based system to map queries to guideline sections.

Explanation

Incorrect: Rule-based systems are too rigid for dynamic queries.

Overall explanation

Using retrievers and LLMs ensures that the system dynamically retrieves and explains answers from medical guidelines.

Question 45

A Generative AI Engineer is tasked with selecting a model from a marketplace to perform entity recognition for legal documents. What factors should they prioritize in the model metadata? **(Choose two)**

1. Fine-tuning capabilities for legal datasets.

Explanation

Correct: Fine-tuning ensures the model aligns with the specific requirements of legal document processing.

2. Performance on entity recognition benchmarks.

Explanation

Correct: High performance on benchmarks ensures the model can accurately identify entities in legal documents.

3. Response generation speed.

Explanation

Incorrect: Response speed is less critical for entity recognition tasks compared to accuracy.

4. The number of training parameters.

Explanation

Incorrect: The parameter count is secondary to task-specific performance.

5. Multilingual capabilities.

Explanation

Incorrect: Multilingual features are unnecessary for monolingual tasks like English legal documents.

Overall explanation

Focusing on entity recognition benchmarks and fine-tuning capabilities ensures the model is effective for legal document tasks.